



NORSK DATA

PRODUCT INFORMATION

Product name

COSMOS X.25 Option for PIOC

Product number

210570A

DESCRIPTION OF PRODUCT

The COSMOS X.25 Option for PIOC provides for access to the public data networks (PDNs). Both the extension of a COSMOS network over the PDN and access to other vendors' machines are made possible. To access other ND hosts (and thus extending the COSMOS network) the COSMOS X.25 Network Server is used. The Network Server is delivered as a part of this product. Other vendors' hosts can be accessed by using the product COSMOS X.29 PAD (ND 210405), and the ND host can be accessed from other vendors' hosts by using the product COSMOS X.29 Server (ND 210578). An OSI Transport Service (ND 211023) is also available.

REASON(S) FOR NEW RELEASE

New revision A06.

Errors corrected and modifications.

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1 PREREQUISITES

The following is required in order to use this product.

1.1 HARDWARE PREREQUISITES

The following hardware must be present for this product to be used:

CPU type (any of the following):

- 100 Standard
- 100 CX
- 110 Standard
- 110 CX
- 120 Standard
- 120 CX
- 500
- 5000

Floating format: *Any*

Other hardware:

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PIOC interface card
with 256K Words of memory.
This product must use PIOC 0 !

108670

1.2 SOFTWARE PREREQUISITES

The following software must be available on the system before the product is installed:

Operating system:

Name

Version/Revision

SINTRAN III - VSX

>= K

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210570AOther software:Product nameProduct numberX-MESSAGE, version K or later, with
a minimum of 13 ports.

210373

BACKUP-SYSTEM, version H or later
(used during installation).

210337

1.3 OTHER PREREQUISITES

Minimum Permanent Mass Storage:

UserSpace (pages)Number of filesCOS-NETSERVERS
RT800
6426
1

	Minimum	Maximum
Number of segments (ND-100)	7	7
Number of RT descriptions	5	5

Space required on segment files: 565 pages

Requirements regarding internal devices:

None

Requirements regarding batch processors:

None

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2 DEPENDENCIES

In addition, the following dependencies exist between this and other products. The hardware and software listed here, may for example be necessary to make full use of the product.

2.1 HARDWARE DEPENDENCIES

None

2.2 SOFTWARE DEPENDENCIES

None

2.3 OTHER DEPENDENCIES

None

3 DOCUMENTATION

Included with the product

COSMOS X.25 Option Operator's Guide.....	30.034.01 EN
COSMOS X.25 Reference Manual.....	60.138.01 EN
PIOC Hardware Ref. Manual ND-865/ND-867...	02.004.01 EN
PIOC Software Guide.....	60.161.03 EN

Optional (must be purchased)

<u>Title</u>	<u>Number</u>
COSMOS X.25 Option Programmer Guide.....	60.227.01 EN

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4 INSTALLATION PROCEDURE

4.1 COPYING THE FILES FROM DISKETTES

In the following instructions, please replace <rev> with the current revision number.

- Log in as user SYSTEM.
- Make available the user COS-NETSERVERS with at least 800 pages and 26 files free. The user X25-NETWORK or any suitable user can also host the product, but be aware that the file X25PI-PROFILE-:SAVE must always be present under either user SYSTEM, COS-NETSERVERS or X25-NETWORK.
- Make available the installation user with at least 200 pages and 5 files free. The user UTILITY is proposed as installation user by the installation program.
- Insert the diskette with the directory name 210570A<rev>-XX-01D.
- Execute these commands (operator input is underlined):

```
@ENTER-DIRECTORY,210570A<rev>-XX-01,FLOPPY-DISC-1,0-  
@(210570A-XX-01:FLOPPY-USER)INST-X25PI-A-
```

The installation procedure then starts. It will:

- Check chosen users.
- Delete old versions of the product.
- Copy the new files to the user hosting the product.
- Prepare the mode-files according to chosen values.

NOTE: Previous versions of the configuration dependant files X25PI-START-A:MODE and X25NS-START-A:MODE will be deleted by the installation program if not saved. If new START files are used, they must be edited before the system can be started. This editing is done automatically by the installation program, but can also be done manually.

4.2 RUNNING WITH SINTRAN < K 3200B

If you run a SINTRAN K revision 3200B or older the file X25PI-SIN3-K-A:PATC must be run. To make patches effective, the system must be restarted. If patches are not run, the X25 module may crash at any time.

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4.3 LOADING

When you have installed the new X.25 system, you can load it using the mode file X25PI-SYSLO-A:MODE. This file should be called from the HENT-MODE:MODE file, which is run at "COLD-START".

4.4 CONFIGURING AND STARTING

The mode-file X25PI-START-A:MODE is used to initialize and start the X.25 packet- and link-level. The parameters in this file should be edited to suit the X.25 connection supplied by your local PDN authority.

The file X25NS-START-A:MODE starts the X.25 Network Server, defines COSMOS X.25 connections, and also calls X25PI-START-A:MODE. The file X25NS-START-A:MODE should be called from your LOAD-MODE:MODE file in order to make COSMOS X.25 start automatically after a "WARM-START".

The file X25PI-OPEREX-A:LIST shows an example of output from the mode-file X25NS-START-A.

4.5 GETTING THE PIOC STATUS WORD

To ease troubleshooting and correction a special program is delivered with this product: X25PI-PISTAT-A<rev>:PROG.

It will give you the size of the PIOC memory, tell you in which memory-banks the PIOC is located, and give its start address. The size of the PIOC should ALWAYS be 512KBytes, while the other data can be used if SINTRAN need to be reconfigured (see below!).

4.6 TROUBLESHOOTING

The most common error when installing the X.25 PIOC Option is that the physical memory in the PIOC is used by SINTRAN for some system table. The error occurs when the START-MODULE command is given in the X25 Command program and results in the error message : "Unable to load PIOC"

If the PIOC physical memory is moved to the middle of the physical address range the problem may be resolved. To do this the system supervisor should consult the "PIOC Hardware Reference Manual".

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If RT-common is set up in the PIOC memory, it can be moved by means of the SINTRAN-SERVICE-PROGRAM:

Ex.

```
@SINTRAN-SERVICE-PROGRAM~
*DEFINE-RTCOMMON-SIZE~
RTCOMMON SIZE: <size>~
FIRST PHYSICAL PAGE(OCT): <first page>~
IMAGE? Y~
SAVE? Y~
```

If the RTCOMMON area is not used, you may just remove it by giving 0 (zero) as the size. You have to do @RESTART-SYSTEM to bring the new placing of the RTCOMMON into effect.

PLEASE NOTE that all RT-programs using the RT-COMMON must be reloaded before they are restarted.

Another method is to reserve the PIOC address space exclusively for (your) user processes:

Ex.

```
@SINTRAN-SERVICE-PROGRAM~
*CHANGE-TABLE~
USER-RESERVED-DEVICE-NUMBERS
USER-RESERVED-MEMORY-AREA
MEMORY-AREA-UNAVAILABLE-FOR-SWAPPING
MEMORY-AREA-INVISIBLE-FOR-THIS-SYSTEM
TABLE: USER-RESERVED-MEMORY-AREA~
FUNCTION: INSERT-ELEMENT~
IMAGE OR SAVE-AREA (DEFAULT IS IMAGE): ~
FIRST PAGE (OCT): <first page of PIOC in physical memory>~
LAST PAGE (OCT): <above number + 400>~
FUNCTION: INSERT-ELEMENT~
IMAGE OR SAVE-AREA (DEFAULT IS IMAGE): SAVE-AREA~
FIRST PAGE (OCT): <first page of PIOC in physical memory>~
LAST PAGE (OCT): <above number + 400>~
FUNCTION: EXIT~
*EXIT
```

<First page of PIOC in physical memory> is obtained from the PIOC status word as described in the previous chapter. The system must be restarted before the reservation becomes active. The above effect can also be obtained by placing the loading and starting of X.25 very early in the LOAD-MODE file (immediately after X-MESSAGE is started).

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4.7 CHANGES IN X-MESSAGE AND FILE-TRANSFER

It is necessary to increase some of the X-MESSAGE network timeouts. You must do the following:

With SINTRAN K:

Set X5T01 to 400 (ie. 40 seconds).
 Set X5T02 to 400 (ie. 40 seconds).
 Set X5NGT to 200 (ie. 20 seconds).

This is done using the XMSG installation program, which changes the parameters of XMSG. You must then reload and restart XMSG. This should be done on all the machines in the COSMOS network.

With SINTRAN L (or later):

Set 'Timeout in XTUs when receiving datagrams' to 400
 Set 'Timeout in XTUs when transmitting datagrams' to 400
 Set 'Gateway timeout in XTUs when sending to net server' to 200

These parameters are changed in the S3-CONFIG program. Afterwards, you must run the command GENERATE in S3-CONFIG, and then COLD-START the machine. This should be done on all the machines in the COSMOS network.

Additionally, the FILE-TRANSFER timeout should be patched to 170B (=120 seconds) on the segment XFTRAD, like this:

```
@LOOK-AT SEGMENT XFTRAD~
YYYYYY/xxx 170~
.-
```

YYYYYY is the location of the timeout value on the XFTRAD segment, and can be found in the load-file COS-XFTRA:MODE.

5 MODIFICATIONS

5.1 NEW FUNCTIONS/COMMANDS

None

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5.2 CHANGED FUNCTIONS/COMMANDS

None

5.3 REMOVED FUNCTIONS/COMMANDS

None

5.4 OTHER MODIFICATIONS

The X.25 Network Server is now able to handle a list of up to 50 DTE definitions. The previous maximum was 40. The limitation of 16 simultaneous connections to different DTEs is still valid.

The X.25 Diagnostic program is enhanced to handle a list of up to 50 DTEs.

6 ERRORS CORRECTED

6.1 INSTALLATION PROGRAM

A previous error in the installation program, which made it impossible to use under SINTRAN L, is now corrected.

6.2 X.25 PIOC CODE

An error in the link level part, which under bad physical link conditions could make the link level go into an endless timer recovery state, is now corrected.

A patch increasing internal buffer capacity, which earlier could be required on systems with heavy X.25 load, is now included in the loading of the product.

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In part 1 of the installation menu, called 'Delete product files of the previous revision', please always answer YES (=default) in response to the following two questions :

'Do you have a previous version of the product installed ?'

and

'Do you want to retain files containing site dependant params ?'

7.2 XMSG-COMMAND PROGRAM ON SINTRAN K

In version A06 of this product, the mode-jobs which call the XMSG-COMMAND program expect to find it on user SYSTEM, which is correct from SINTRAN L. On SINTRAN K, however, the XMSG-COMMAND program lies on the user UTILITY. Therefore, if you run SINTRAN K : dump XMSG-COMMAND reentrant, or copy it to user SYSTEM also.

7.3 PACKET SIZE

The Operator Guide indicates that it is possible to define a packet size bigger than 128 bytes in the X.25 profile. This is NOT possible.

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8 MISCELLANEOUS

Here, inconsistencies between this implementation and the currently available documentation are mentioned.

8.1 DEMONSTRATION PROGRAMS

Two demonstration programs are included in this product. They can be used to test whether X.25 is properly installed. The demonstration programs assume that X.25 runs on the system called 'X25PTEST'. Hence, before you run the programs, you must do this:

Log in as user SYSTEM

@XMSG-COMMAND-

X-C: DEFINE-REMOTE-NAME,,X25PTEST,<local system number>-

X-C: EXIT-

To be able to run in link level loopback mode it is necessary to generate internal clock rate. This is done by patching the location LOCALCL on the segment X25SUPV to 1 before the CHANGE-PROFILE command is executed.

You must use the CHANGE-PROFILE command in the X25-COMMAND program to change X.25 to run in link level loopback mode. Your DTE number must be changed to '1234'. The COSMOS X.25 Operator Guide explains how this is done.

Then you can start the demonstration programs. Log in as user COS-NETSERVERS. The listening program (X25P-DEMO-L) must be started first. When it says it has attached to X.25, you can start the initiator (X25P-DEMO-I) on another terminal.

Example of use:

The listening program must be started first:

@X25P-DEMO-L-

Program attached to X.25 and waiting for an incoming call

Connection established

Message received: Hello - how are you?

Message sent: Very well thank you!

Disconnect indication received

Demonstration program successfully completed

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The initiator must be started after the listener:

@X25P-DEMO-I-1

Connect request sent

Connection established

Message sent:Hello - how are you?

Message received:Very well thank you!

Disconnect sent

Demonstration program successfully completed

8.2 CHANNEL GROUP NUMBER

Some PDNs (e.g. PSS in the UK) use the channel group number in addition to the channel number. When you apply for an X.25 connection, you must specify the number of channels you want to use. The network provider may tell you which channel group they belong to. The X.25 profile command in the command program requires a channel number only. Hence, you must transform the channel group number and channel number to a channel number for the command program. It is done in this way:

Lower bound: Multiply the channel group number by 256.

Upper bound: Lower bound + (number of channel -1)

Example:

The network provider gives you this channel information:

- * Channel group is 4.

- * Number of channels is 8.

The channel range used in the command program will then be:

- * Lower bound: $(4 \times 256) = 1024$ (default value)

- * Upper bound: $(4 \times 256) + 7 = 1031$ (default value)

8.3 DIAGNOSTIC PROGRAM

The X.25 diagnostic program (X25PI-DIAG-A:PROG) is a tool, which enables an operator to establish a fast and simple diagnostic on the current status of connections within a COSMOS X.25 network.

In the initialization phase the program scans the X.25 start-up file (COS-NETSERVERS)X25NS-START-A:MODE for network connection definitions, and makes a list of DTE names and corresponding addresses. If no DTE definitions are found, the program will prompt for an alternative file to scan.

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Within the program you can check the communication with one, or all of the DTE entries in the list. It is also possible to define additional DTEs in the program. The command HELP will list the available options.

8.4 NETWORK SERVER TIMEOUTS

The network server checks all the connections in use. If it takes too long time to complete a connect request, the connect is aborted, and the user data returned to X-MESSAGE. The connect request timeout is set to 60 seconds.

The network server will clear a connection if it takes too long to receive the acknowledgement of a sent data packet. The data packet timeout is set to 90 seconds.

8.5 NETWORK SERVER AND PACKET LEVEL WINDOW SIZE

The network server will try to negotiate a bigger packet level window size when a CALL REQUEST is sent. The window negotiation can be switched off with the variable X25EXTF.

Values: 0 - No negotiation will be attempted.
1 - Negotiation (default).

8.6 NETWORK SERVER LOGGING

Using the log option it is possible to log messages exchanged with the packet level and X-MESSAGE (i.e. XGATE). The log option can be switched on/off while the network server is running. The logged information is written on the file COS-X25NS:LOGS on user RT. This file must be created before you can turn the log on. The network server will count the number of entries written on the file. When this number exceeds a pre-defined maximum, the file is rewound. You need approximately 30 pages for the default number of entries (=10000). Three patch variables are used for the operation of the log option. The layout of the patch variables is:

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LOCATION	FUNCTION	POSSIBLE VALUES
LGPRF	Select log items (bit map)	Bit 0 - Log msg to XGATE. Bit 1 - Log msg from XGATE. Bit 2 - Log msg to X.25. Bit 3 - Log msg from X.25. Bit 4 - Log msg X.25 errors only. Bit 5 - Log only first 30 byte of data msg (in combination with bit 0,1,2,3). Bit 6 - Log X.25 Resets and Disconnects with Cause >< 0.
LGPRF+1	Change log state	0 - Initial value. 1 - Turn log on. 2 - Turn log off. 3 - Unlock log file.
LGPRF+2	Max number of entries in log file	1750B - Initial value (= 1000D). xxxxB - Choose desired value (remember to adjust the file size accordingly).

8.7 NETWORK SERVER COMMANDS

The following commands concerning the X.25 Network Server can be used in the XMSG-COMMAND program :

DEFINE-NETWORK-CONNECTION can be used to remove network connections, in addition to defining new ones (just answer with an empty DTE address).

LIST-VERSION is implemented.

LIST-NETWORK-REMOTE-ENDPOINTS is implemented, and gives a list of the defined network connections, with additional information on the current state of the connections.

8.8 NSAP ADDRESSING

When you use the command DEFINE-NETWORK-CONNECTION in the XMSG-COMMAND program, one parameter indicates the DTE address. This parameter is changed. It can be used to indicate more than just the DTE number. You can specify an NSAP (network service access point). You need XMSG version K (or later) if you want to use this feature.

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The parameter must be structured if you want to use it as an NSAP parameter. If you want to send a DTE subaddress on outgoing calls, and ignore it when finding a match for incoming calls, the subaddress is delimited with a '.'.

Example:

2422110006.55 The network server will send the DTE number 242211000655, but for incoming calls search for a match with the number 2422110006.

If you do not want to ignore the subaddress for incoming calls, you do not need the delimiter (incoming address = outgoing address).

The X.25 call user data field can be used for extended addressing. The delimiter '+' is used to delimit the DTE address from the address part in the call user data field. The address used for outgoing calls must be indicated first. After the delimiter '/' the address for incoming calls is given. If they are identical, they must be entered twice. Usually they are not. Since this NSAP parameter is one parameter in the XMSG command program, spaces and commas are not allowed in the parameter string. The ';' is interpreted as ',', and '\$' as ' ' (space). ';;' must be used to indicate ';' and '\$\$' for '\$'. '//' is interpreted as '/' (no delimiter). A '/' as the last character in the outgoing address part is impossible.

Example:

40+PSS(AAA;BBB).NFS__63500140/GB__263500140
 Outgoing call user data field: PSS(AAA,BBB).NFS__63500140
 Incoming call user data field: GB__263500140

...../	is interpreted as		(redundant delimiter)
.....//		/	(character '/', no delimiter no incoming address part)
..//.		../.	(character '/', no delimiter)
..//..		..D..	(Delimiter)
..//..		..D/.	(Delimiter, first incoming character is '/')